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References:

1. R. Gaonkar, Microprocessor Architecture Programming, Penram International Publishing
2. Hall D.V., microprocessor & Interfacing-Programming & hardware, TMH, 2008
3.https://www.tutorialspoint.com/microprocessor/microprocessor_intel_8253_programmable_interval_timer.htm.

Faculty HOD Dean Academics

UNIT-3 (Interfacing with I/O devices)

Interfacing Concept & Interfacing of I/O Devices

Lecture No.-1

Interface is the path for communication between two components. Interfacing is of two types, memory interfacing and I/O interfacing.

Input-Output Interface

It is used as an method which helps in transferring of information between the internal storage devices i.e. memory and the external peripheral device . A peripheral device is that which provide input and output for the computer, it is also called Input-Output devices.

Example: A keyboard and mouse provide Input to the computer are called input devices.

A monitor and printer that provide output to the computer are called output devices.

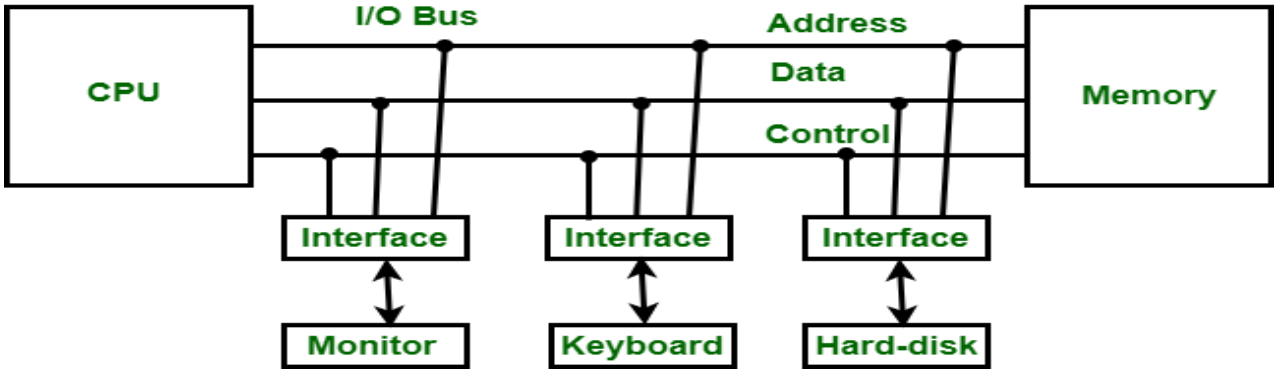


Fig: Input-Output Interface

8085 Interfacing Pins

Following is the list of 8085 pins used for interfacing with other devices −

- A_{15} - A_8 (Higher Address Bus)
- AD_7 - AD_0(Lower Address/Data Bus)
- ALE
- RD
- WR
- READY

Ways of Communication – Microprocessor with the Outside World

There are two ways of communication in which the microprocessor can connect with the outside world.

- Serial Communication Interface
- Parallel Communication interface

Serial Communication Interface − In this type of communication, the interface gets a single byte of data from the microprocessor and sends it bit by bit to the other system serially and vice-a-versa.

Parallel Communication Interface – In this type of communication, the interface gets a byte of data from the microprocessor and sends it bit by bit to the other systems in simultaneous (or) parallel fashion and vice-a-versa.

UNIT-3 (Interfacing with I/O devices)

Interrupt in 8085 MP

Lecture No.-2

Introduction

In the 8085 microprocessor, an interrupt is a signal that temporarily suspends the normal execution of a program and redirects the control to a specific interrupt service routine (ISR). Interrupts allow the microprocessor to respond to external events, such as user input, system events, or hardware signals, without the need for constant polling.

There are five interrupt signals in the 8085 microprocessor:

1. **TRAP:** The TRAP interrupt is a non-maskable interrupt that is generated by an external device, such as a power failure or a hardware malfunction. The TRAP interrupt has the highest priority and cannot be disabled.
2. **RST 7.5:** The RST 7.5 interrupt is a maskable interrupt that is generated by a software instruction. It has the second highest priority.
3. **RST 6.5:** The RST 6.5 interrupt is a maskable interrupt that is generated by a software instruction. It has the third highest priority.
4. **RST 5.5:** The RST 5.5 interrupt is a maskable interrupt that is generated by a software instruction. It has the fourth highest priority.
5. **INTR:** The INTR interrupt is a maskable interrupt that is generated by an external device, such as a keyboard or a mouse. It has the lowest priority and can be disabled.

When microprocessor receives any interrupt signal from peripheral(s) which are requesting its services, it stops its current execution and program control is

transferred to a sub-routine by generating **CALL** signal and after executing sub-routine by generating **RET** signal again program control is transferred to main program from where it had stopped. When microprocessor receives interrupt signals, it sends an acknowledgement (INTA) to the peripheral which is requesting for its service.

Interrupts can be classified into various categories based on different parameters:

1. **Hardware and Software Interrupts –** When microprocessors receive interrupt signals through pins (hardware) of microprocessor, they are known as *Hardware Interrupts*. There are 5 Hardware Interrupts in 8085 microprocessor. They are – *INTR, RST 7.5, RST 6.5, RST 5.5, TRAP Software Interrupts* are those which are inserted in between the program which means these are mnemonics of microprocessor. There are 8 software interrupts in 8085 microprocessor. They are – *RST 0, RST 1, RST 2, RST 3, RST 4, RST 5, RST 6, RST 7*.
2. **Vectored and Non-Vectored Interrupts –** *Vectored Interrupts* are those which have fixed vector address (starting address of sub-routine) and after executing these, program control is transferred to that address. Vector Addresses are calculated by the formula 8 * TYPE

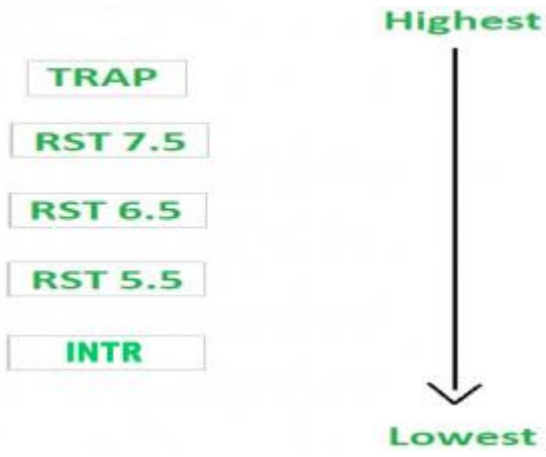
INTERRUPT	VECTOR ADDRESS
TRAP (RST 4.5)	24 H
RST 5.5	2C H
RST 6.5	34 H
RST 7.5	3C H

1. For Software interrupts vector addresses are given by:

INTERRUPT	VECTOR ADDRESS
RST 0	00 H
RST 1	08 H
RST 2	10 H
RST 3	18 H
RST 4	20 H
RST 5	28 H
RST 6	30 H
RST 7	38 H

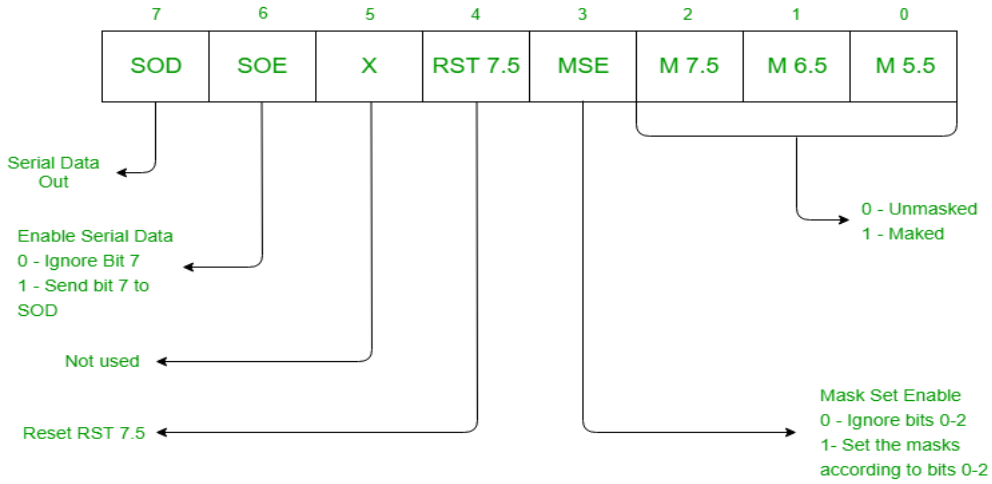
1. ***Non-Vectored Interrupts***- are those in which vector address is not predefined. The interrupting device gives the address of sub-routine for these interrupts. *INTR* is the only non-vectored interrupt in 8085 microprocessor.
2. **Maskable and Non-Maskable Interrupts** – *Maskable Interrupts* are those which can be disabled or ignored by the microprocessor. These interrupts are either edge-triggered or level-triggered, so they can be disabled. *INTR, RST 7.5, RST 6.5, RST 5.5* are maskable interrupts in 8085 microprocessor. Non-Maskable Interrupts are those which cannot be disabled or ignored by microprocessor. *TRAP* is a non-maskable interrupt. It consists of both level as well as edge triggering and is used in critical power failure conditions.

Priority of Interrupts – When microprocessor receives multiple interrupt requests simultaneously, it will execute the interrupt service request (ISR) according to the priority of the interrupts.

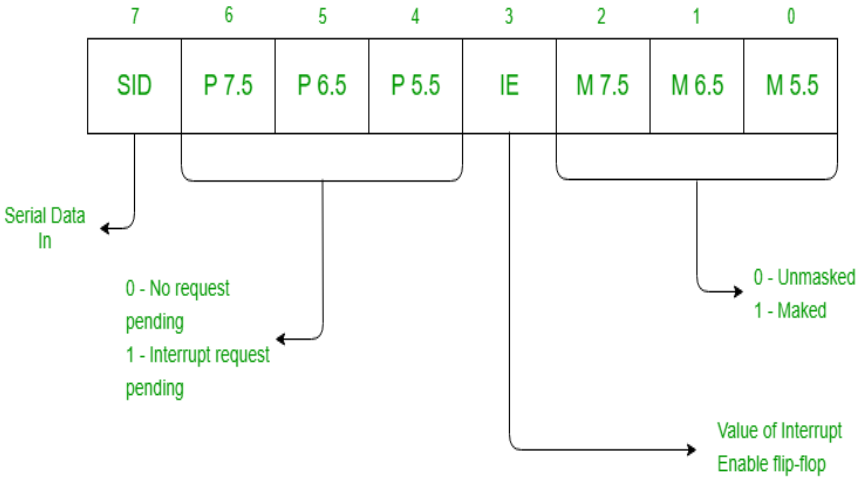


Instruction for Interrupts

1. **Enable Interrupt (EI) –** The interrupt enable flip-flop is set and all interrupts are enabled following the execution of next instruction followed by EI. No flags are affected. After a system reset, the interrupt enable flip-flop is reset, thus disabling the interrupts. This instruction is necessary to enable the interrupts again (except TRAP).
2. **Disable Interrupt (DI) –** This instruction is used to reset the value of enable flip-flop hence disabling all the interrupts. No flags are affected by this instruction.
3. **Set Interrupt Mask (SIM) –** It is used to implement the hardware interrupts (RST 7.5, RST 6.5, RST 5.5) by setting various bits to form masks or generate output data via the Serial Output Data (SOD) line. First the required value is loaded in accumulator then SIM will take the bit pattern from it.



Read Interrupt Mask (RIM) – This instruction is used to read the status of the hardware interrupts (RST 7.5, RST 6.5, RST 5.5) by loading into the A register a byte which defines the condition of the mask bits for the interrupts. It also reads the condition of SID (Serial Input Data) bit on the microprocessor.



UNIT-3 (Interfacing with I/O devices)

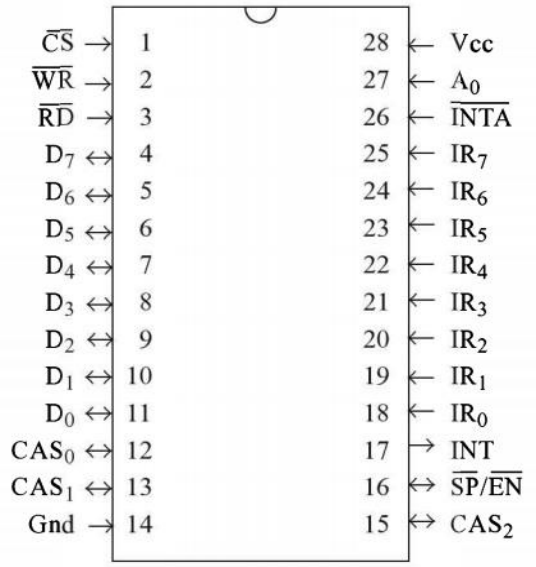
Programmable Interrupt Controller (8259A)

Lecture No.-3

The 8259 is known as the Programmable Interrupt Controller (PIC) microprocessor. In 8085 and 8086 there are five hardware interrupts and two hardware interrupts respectively. Bu adding 8259, we can increase the interrupt handling capability. This chip combines the multi-interrupt input source to single interrupt output. This provides 8-interrupts from IR0 to IR7. Let us see some features of this microprocessor.

Features:

- This chip is designed for 8085 and 8086.
- It can be programmed either in edge triggered, or in level triggered mode
- We can mask individual bits of Interrupt Request Register.
- By cascading 8259 chips, we can increase interrupts up to 64 interrupt lines
- Clock cycle is not needed.



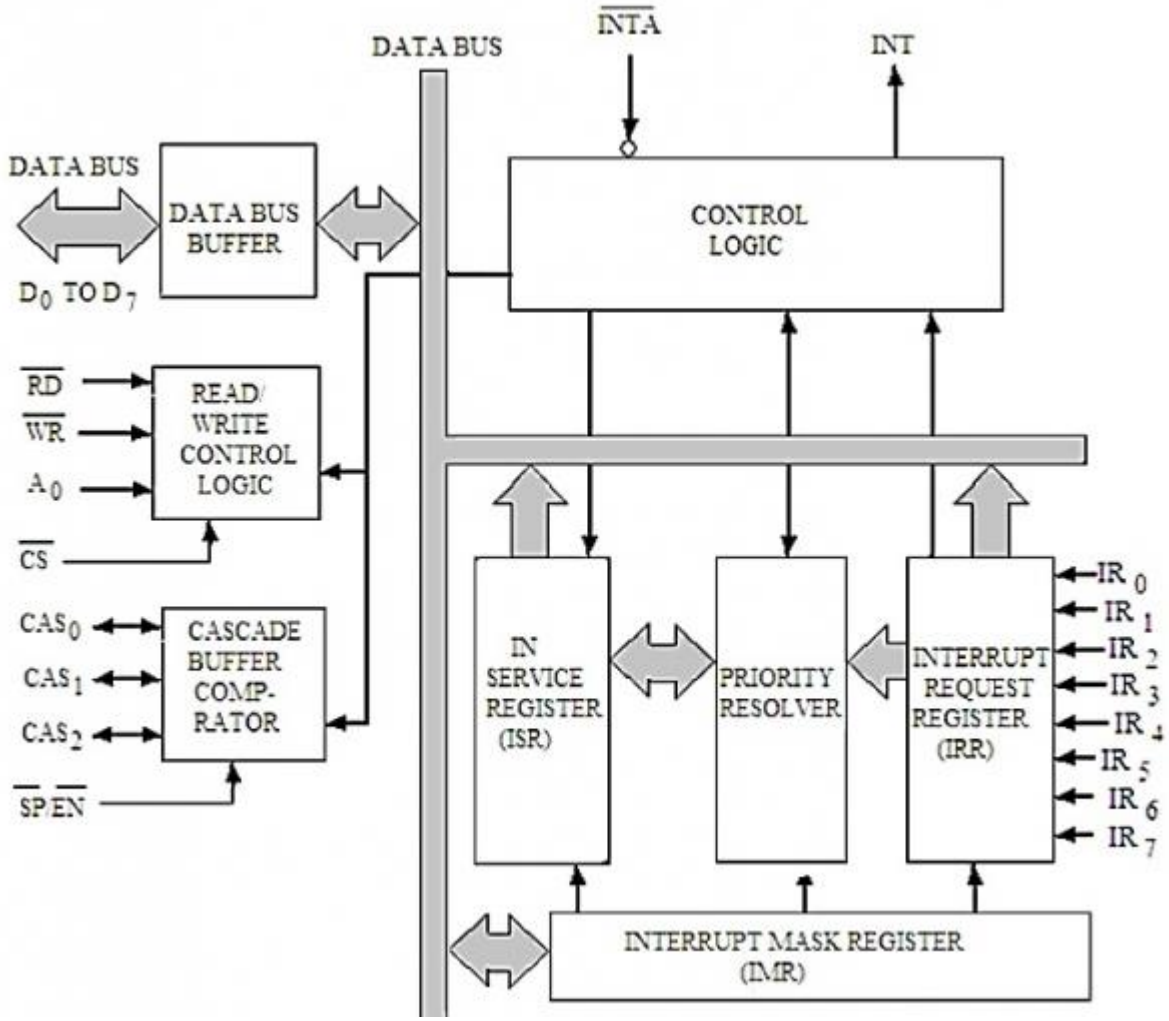


Fig: Block diagram of 8259

Description of block diagram

Block	Description
Data Bus Buffer	This block is used to communicate between 8259 and 8085/8086 by acting as buffer. It takes the control word from 8085/8086 and sends it to the 8259. It transfers the op code of the selected interrupts and address of ISR to the other connected microprocessor. It can send maximum 8-bit at a time.
R/W Control Logic	This block works when the value of pin CS is 0. This block is used to flow the data depending upon the inputs of RD and WR. These are active low pins for read and write.
Control Logic	It controls the functionality of each block. It has pin called INTR. This is connected to other microprocessors for taking the interrupt request. The INT pin is used to give the output. If 8259 is enabled, and also the interrupt flags of other microprocessors are high then this causes the value of the output INT pin high, and in this way this chip can responds requests made by other microprocessors.
Interrupt Request Register	It stores all interrupt level that are requesting for interrupt service.
Interrupt Service Register	It stores interrupt level that are currently being execute.
Interrupt Mask Register	It stores interrupt level that will be masked, by storing the masking bits of interrupt level.
Priority Resolver	It checks all three registers, and set the priority of the interrupts. Interrupt with the highest priority is set in the ISR register. It also reset the interrupt level which is already been serviced in the IRR.
Cascade Buffer	To increase number of interrupt pin, we can cascade more number of pins, by using cascade buffer. When we are going to increase the interrupt capability, CSA lines are used to control multiple interrupts.

UNIT-3 (Interfacing with I/O devices)

Programmable Peripheral Interface (8255A)

Lecture No.-4

The 8255A is a general purpose programmable I/O device designed to transfer the data from I/O to interrupt I/O under certain conditions as required. It can be used with almost any microprocessor.

It consists of three 8-bit bidirectional I/O ports (24I/O lines) which can be configured as per the requirement.

Ports of 8255A

8255A has three ports, i.e., PORT A, PORT B, and PORT C.

- **Port A** contains one 8-bit output latch/buffer and one 8-bit input buffer.
- **Port B** is similar to PORT A.
- **Port C** can be split into two parts, i.e. PORT C lower (PC0-PC3) and PORT C upper (PC7-PC4) by the control word.

These three ports are further divided into two groups, i.e. Group A includes PORT A and upper PORT C. Group B includes PORT B and lower PORT C. These two groups can be programmed in three different modes, i.e. the first mode is named as mode 0, the second mode is named as Mode 1 and the third mode is named as Mode 2.

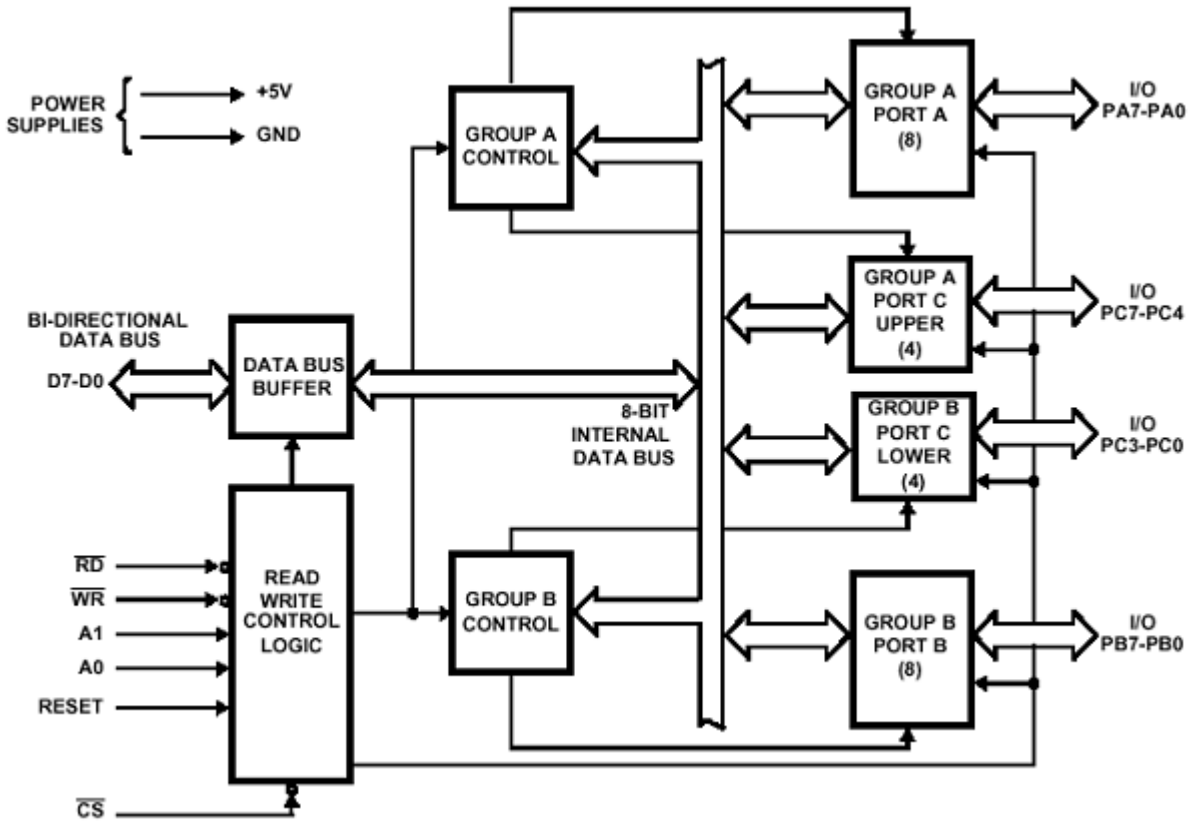
Features of 8255A

The prominent features of 8255A are as follows −

- It consists of 3 8-bit IO ports i.e. PA, PB, and PC.
- Address/data bus must be externally demux'd.
- It is TTL compatible.
- It has improved DC driving capability.

8255 Architecture

The following figure shows the architecture of 8255A



UNIT-3 (Interfacing with I/O devices)

DMA Controller (8257)

Lecture No.-5

DMA stands for Direct Memory Access. It is designed by Intel to transfer data at the fastest rate. It allows the device to transfer the data directly to/from memory without any interference of the CPU.

Using a DMA controller, the device requests the CPU to hold its data, address and control bus, so the device is free to transfer data directly to/from the memory. The DMA data transfer is initiated only after receiving HLDA signal from the CPU.

DMA Operations

Following is the sequence of operations performed by a DMA −

- Initially, when any device has to send data between the device and the memory, the device has to send DMA request (DRQ) to DMA controller.
- The DMA controller sends Hold request (HRQ) to the CPU and waits for the CPU to assert the HLDA.
- Then the microprocessor tri-states all the data bus, address bus, and control bus. The CPU leaves the control over bus and acknowledges the HOLD request through HLDA signal.

- Now the CPU is in HOLD state and the DMA controller has to manage the operations over buses between the CPU, memory, and I/O devices.

Features of 8257

Here is a list of some of the prominent features of 8257 −

- It has four channels which can be used over four I/O devices.
- Each channel has 16-bit address and 14-bit counter.
- Each channel can transfer data up to 64kb.
- Each channel can be programmed independently.
- Each channel can perform read transfer, write transfer and verify transfer operations.
- It generates MARK signal to the peripheral device that 128 bytes have been transferred.
- It requires a single phase clock.
- Its frequency ranges from 250Hz to 3MHz.
- It operates in 2 modes, i.e., Master mode and Slave mode.

8257 Architecture

The following image shows the architecture of 8257

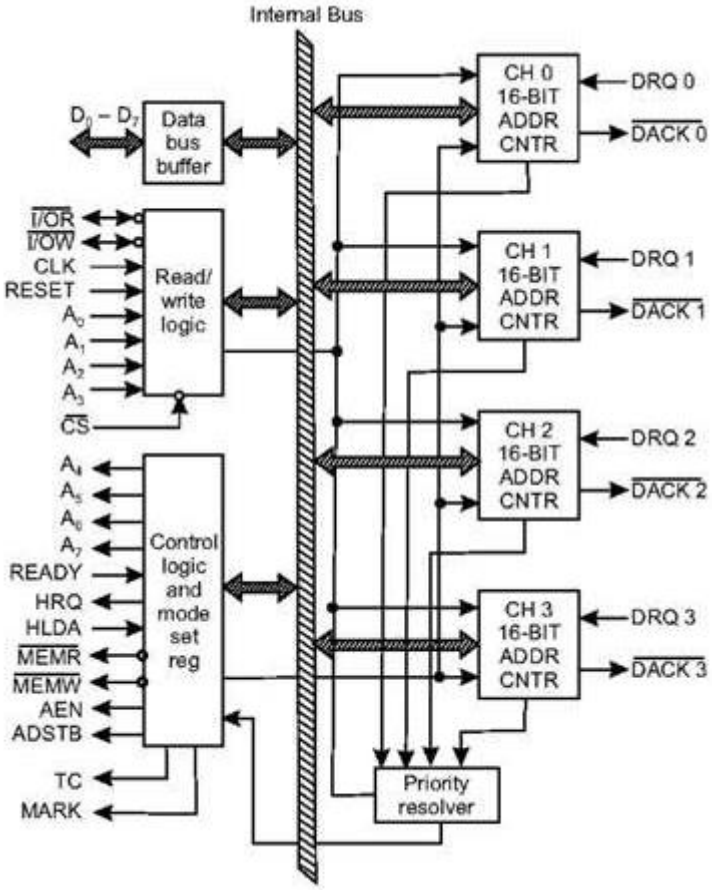
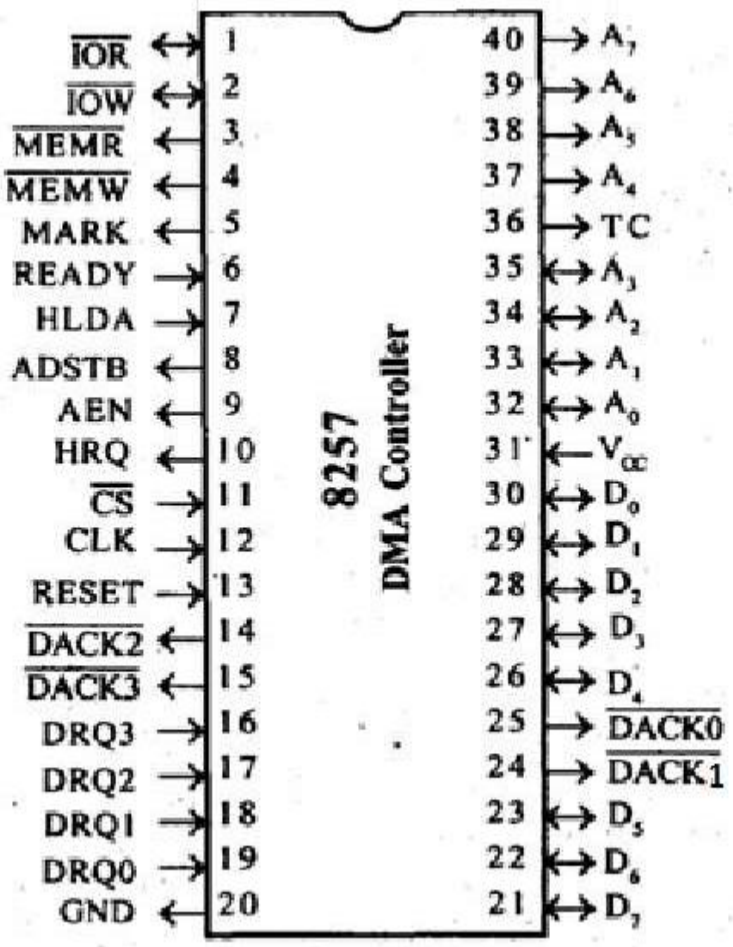


Fig.: Architecture of 8057

8257 Pin Description

The following image shows the pin diagram of a 8257 DMA controller:



DRQ_0–DRQ**3**

These are the four individual channel DMA request inputs, which are used by the peripheral devices for using DMA services. When the fixed priority mode is selected, then DRQ_0 has the highest priority and DRQ_3 has the lowest priority among them.

$DACK_0$ – $DACK_3$

These are the active-low DMA acknowledge lines, which updates the requesting peripheral about the status of their request by the CPU. These lines can also act as strobe lines for the requesting devices.

$D_o - D_7$

These are bidirectional, data lines which are used to interface the system bus with the internal data bus of DMA controller. In the Slave mode, it carries command words to 8257 and status word from 8257. In the master mode, these lines are used to send higher byte of the generated address to the latch. This address is further latched using ADSTB signal.

IOR

It is an active-low bidirectional tri-state input line, which is used by the CPU to read internal registers of 8257 in the Slave mode. In the master mode, it is used to read data from the peripheral devices during a memory write cycle.

IOW

It is an active low bi-direction tri-state line, which is used to load the contents of the data bus to the 8-bit mode register or upper/lower byte of a 16-bit DMA address register or terminal count register. In the master mode, it is used to load the data to the peripheral devices during DMA memory read cycle.

CLK

It is a clock frequency signal which is required for the internal operation of 8257.

RESET

This signal is used to RESET the DMA controller by disabling all the DMA channels.

A_o - A_3

These are the four least significant address lines. In the slave mode, they act as an input, which selects one of the registers to be read or written. In the master mode, they are the four least significant memory address output lines generated by 8257.

CS

It is an active-low chip select line. In the Slave mode, it enables the read/write operations to/from 8257. In the master mode, it disables the read/write operations to/from 8257.

A_4 - A_7

These are the higher nibble of the lower byte address generated by DMA in the master mode.

READY

It is an active-high asynchronous input signal, which makes DMA ready by inserting wait states.

HRQ

This signal is used to receive the hold request signal from the output device. In the slave mode, it is connected with a DRQ input line 8257. In Master mode, it is connected with HOLD input of the CPU.

HLDA

It is the hold acknowledgement signal which indicates the DMA controller that the bus has been granted to the requesting peripheral by the CPU when it is set to 1.

MEMR

It is the low memory read signal, which is used to read the data from the addressed memory locations during DMA read cycles.

MEMW

It is the active-low three state signal which is used to write the data to the addressed memory location during DMA write operation.

ADST

This signal is used to convert the higher byte of the memory address generated by the DMA controller into the latches.

AEN

This signal is used to disable the address bus/data bus.

TC

It stands for 'Terminal Count', which indicates the present DMA cycle to the present peripheral devices.

MARK

The mark will be activated after each 128 cycles or integral multiples of it from the beginning. It indicates the current DMA cycle is the 128th cycle since the previous MARK output to the selected peripheral device.

V_{cc}

It is the power signal which is required for the operation of the circuit.

UNIT-3 (Interfacing with I/O devices)

Programmable Interval Timer 8253/8254

Lecture No.-6

The Intel 8253 and 8254 are Programmable Interval Timers (PTIs) designed for microprocessors to perform timing and counting functions using three 16-bit registers. Each counter has 2 input pins, i.e. Clock & Gate, and 1 pin for "OUT" output. To operate a counter, a 16-bit count is loaded in its register. On command, it begins to decrement the count until it reaches 0, then it generates a pulse that can be used to interrupt the CPU.

Difference between 8253 and 8254

The following table differentiates the features of 8253 and 8254:

8253	8254
Its operating frequency is 0 - 2.6 MHz	Its operating frequency is 0 - 10 MHz
It uses N-MOS technology	It uses H-MOS technology
Read-Back command is not available	Read-Back command is available
Reads and writes of the same counter cannot be interleaved.	Reads and writes of the same counter can be interleaved.

Features of 8253 / 54

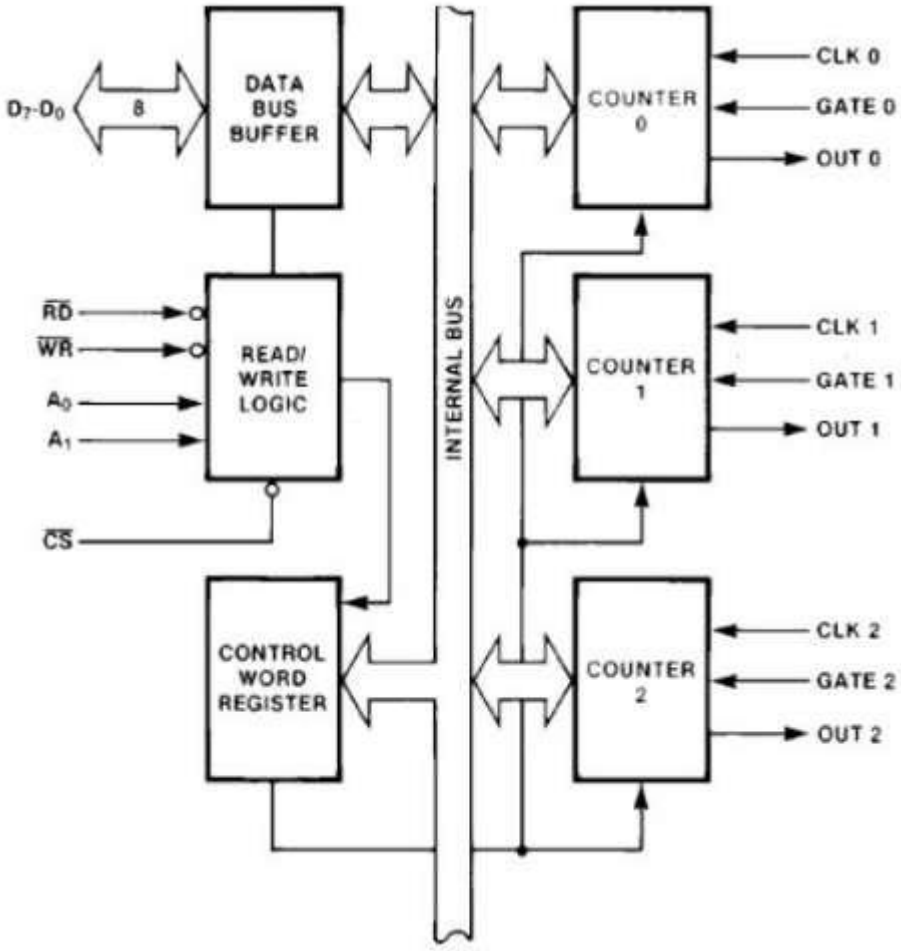
The most prominent features of 8253/54 are as follows –

- It has three independent 16-bit down counters.
- It can handle inputs from DC to 10 MHz.
- These three counters can be programmed for either binary or BCD count.
- It is compatible with almost all microprocessors.

- 8254 has a powerful command called READ BACK command, which allows the user to check the count value, the programmed mode, the current mode, and the current status of the counter.

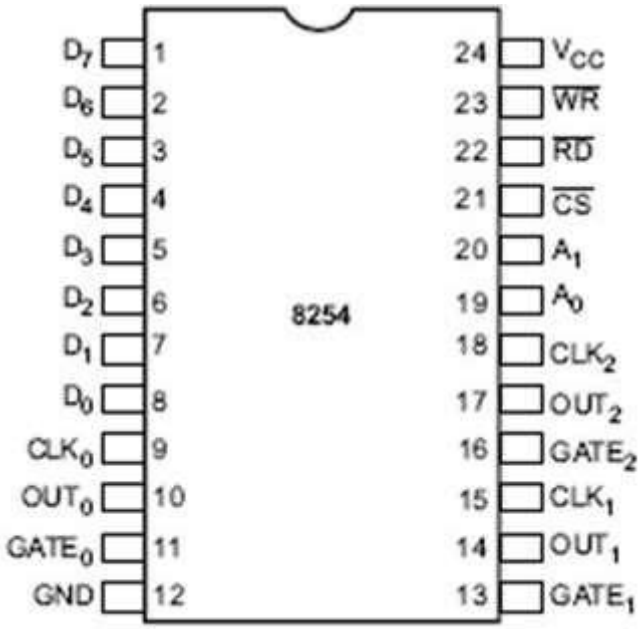
8254 Architecture

The architecture of 8254 looks as follows:



8254 Pin Description

Here is the pin diagram of 8254:



In the above figure, there are three counters, a data bus buffer, Read/Write control logic, and a control register. Each counter has two input signals - CLOCK & GATE, and one output signal - OUT.

Data Bus Buffer

It is a tri-state, bi-directional, 8-bit buffer, which is used to interface the 8253/54 to the system data bus. It has three basic functions −

- Programming the modes of 8253/54.
- Loading the count registers.
- Reading the count values.

Read/Write Logic

It includes 5 signals, i.e. RD, WR, CS, and the address lines A_0 & A_1. In the peripheral I/O mode, the RD and WR signals are connected to IOR and IOW,

respectively. In the memory mapped I/O mode, these are connected to MEMR and MEMW.

Address lines A_0 & A_1 of the CPU are connected to lines A_0 and A_1 of the 8253/54, and CS is tied to a decoded address. The control word register and counters are selected according to the signals on lines A_0 & A_1.

A1	A0	Result
0	0	Counter 0
0	1	Counter 1
1	0	Counter 2
1	1	Control Word Register
X	X	No Selection

Control Word Register

This register is accessed when lines A_0 & A_1 are at logic 1. It is used to write a command word, which specifies the counter to be used, its mode, and either a read or write operation. Following table shows the result for various control inputs.

A1	A0	RD	WR	CS	Result
0	0	1	0	0	Write Counter 0
0	1	1	0	0	Write Counter 1
1	0	1	0	0	Write Counter 2
1	1	1	0	0	Write Control Word
0	0	0	1	0	Read Counter 0
0	1	0	1	0	Read Counter 1
1	0	0	1	0	Read Counter 2
1	1	0	1	0	No operation
X	X	1	1	0	No operation
X	X	X	X	1	No operation

Counters

Each counter consists of a single, 16 bit-down counter, which can be operated in either binary or BCD. Its input and output is configured by the selection of modes stored in the control word register. The programmer can read the contents of any of the three counters without disturbing the actual count in process.